

Cycle-Envelope Kernels for Disk Completion

Exact 2^λ compression beyond frontier width

A constructive continuation toward quasi-polynomial unknot recognition

Research contribution prepared for the ProveIt project

Mathematical development and reference software prepared with ChatGPT

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Abstract

Recent ProveIt reports prove an optimal 2^{r-1} representative bound on r labelled boundary arcs when every cap partition is permitted. We strengthen this interface theorem using certified restrictions on both sides. Let past partitions refine σ and future partitions refine ρ . Their bipartite incidence multigraph has one edge per arc. If it is connected and has cycle rank $\lambda = r + 1 - |\sigma| - |\rho|$, the restricted disk-completion matrix has binary rank exactly 2^λ ; its grade- j block has rank $\binom{\lambda}{j}$. These bounds are sharp for every connected envelope, even for arbitrary representative-subset methods. Vertex-splitting equations on the cycle space give an explicit factorization and independently checkable weighted pruning certificates.

For a supplied finite layered disk-patch grammar, safe envelopes are constructible in polynomial time by joining all local options. Repeated cycle-basis reduction preserves minimum-cost successful assemblies. A straightforward algorithm runs in $\text{poly}(I, B)2^{3\lambda_*}$ bit time, where I is explicit grammar length, B bounds cost bit lengths, and λ_* is maximum envelope cycle rank. The sufficient quasi-polynomial regime is therefore $\lambda_* = O(\log^2 n)$, not necessarily $r = O(\log^2 n)$; actual arc counts can remain polynomially large.

The package includes a complete finite-grammar solver, separate pruning and run checkers, and literal triangulated-surface replay. Forty-two test methods pass. Additional audits check 633,531 compatibility entries, 1,000 weighted families, 600 three-way grammar comparisons, and 1,412 fully enumerated assignments. Restricted-envelope benchmarks retain four candidates instead of 16–29 in the unrestricted basis and improve its measured median search times by 1.49–3.04 times. Single-cap preprocessing remains slower than direct scanning. These are abstract-grammar measurements, not native recognizer timings. No complete low-cycle-envelope producer for all knot exteriors, or general quasi-polynomial recognition bound, is established here.

Status. Self-contained proofs and executed standalone software. Classical rank-based and exterior-algebra methods are credited; worldwide priority for equivalent formulations is not asserted. No proof-assistant verification or native runtime integration is claimed.

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1 Research position and the precise advance

1.1 The current repository already has a source bridge

The reviewed ProveIt revision is

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The maintained `Topology/UnknotRecognition` material distinguishes its exact recognizers from the general quasi-polynomial implementation target [1]. The canonical diagram-to-exterior bridge is no longer absent. The inspected source `synthesis/diagram_exterior.tex` constructs and separately checks a compact knot-exterior triangulation from a validated planar one-component PD code. Its pulling subdivision has $20 \max(1, n)$ tetrahedra and $8 \max(1, n)$ boundary triangles for an n -crossing diagram [2]. That source theorem is existing infrastructure, not a result of this delivery.

Recent project reports also describe source-bound normal-surface component analysis and compressing-disk tests for supplied normal vectors [3, 5]. Fast verification of one binary-encoded normal surface does not by itself construct a complete collection of candidate surfaces, expose manageable interfaces, or bound all geometric search choices. Our contribution addresses a search-table parameter after an explicit surface-assembly language has been supplied.

Lackenby’s July 2026 hierarchy paper supplies geometric context. Its Section 9 separates the number of certain algorithmic steps from a running-time estimate for operations whose implementation costs are not yet specified there [10]. We do not obtain an end-to-end bound by treating those operations as unit cost. The conditional application below charges construction, type information, bit lengths, and witness verification.

1.2 What the incoming reports already prove

The incoming *Disk-Completion Kernels* and *Optimal Disc-Completion Bases* manuscripts establish sharp unrestricted binary rank 2^{r-1} and the corresponding weighted representative bound [3, 4]. The signed-continuation manuscript studies orientation consistency and finite patch selection [5]. We inspected Library texts and the incoming archive inventory, but did not unpack and establish byte identity with every incoming ZIP. The source audit records this limitation.

The unrestricted bound cannot be improved while preserving every cap on r arcs. That is not the same as saying every actual continuation language requires that many representatives. A future region may never join certain pairs of arcs. A past region can have analogous restrictions. We ask:

Given certified coarse connectivity envelopes on both sides, what is the exact rank and representative size for all refinements of those envelopes?

We answer this for every pair of envelopes. The exponent $r - 1$ becomes the cycle rank of their incidence graph. We also construct safe envelopes from a complete explicit patch grammar in polynomial time, without first enumerating all suffix states.

1.3 Established methods and the novelty boundary

Cost-ordered rank bases for weighted connectivity problems are classical [6]. Exterior-algebra representative families for linear matroids are also established tools [7]. Gaussian elimination,

the representative replacement argument, and single-exponential connectivity dynamic programming are not claimed as new. The overlap dimension of representable subspaces relates to the standard matroid connectivity function [9]; theorem 3.7 makes this explicit here.

The developed contribution is the exact *two-sided disk-completion* theorem, its sharpness for each connected envelope, the vertex-splitting factorization, the constructive envelope compiler and composition proof, and an independently checked executable search. This is an application and refinement of classical methods. An exhaustive worldwide priority search for every equivalent formulation has not been completed.

2 Topology and the two-sided interface

2.1 The boundary-arc contract

All local surfaces are compact finite PL surfaces, each component having nonempty boundary. An attachment arc is a proper closed interval in a boundary circle. Distinct arcs in one surface are disjoint, including their endpoints, and have disjoint collars. Attachments glue specified arc pairs by homeomorphisms with no other identifications; corners are smoothed. The quotient is again a compact surface with boundary.

One label denotes one actual connected interval, not a point-incidence histogram or an arbitrarily large bundle of parallel intervals. All actual intervals contribute to the Euler calculation. Ambient positions, boundary framings, quadrilateral restrictions, and other continuation-visible geometry must be fixed in a separate type. A partition alone does not certify an embedding in a three-manifold.

Write $c(S) = \text{comp}(S)$ and $\delta(S) = c(S) - \chi(S)$. For a connected orientable surface of genus g with $h \geq 1$ boundary circles, $\delta = 2g + h - 1$. For a connected nonorientable surface with k crosscaps and $h \geq 1$ boundary circles, $\delta = k + h - 1$. Thus $\delta(S) \geq 0$, with equality exactly for disk unions.

Lemma 2.1 (Defect under separated boundary-arc attachments). *Glue r arc pairs between surfaces S, T satisfying the contract. Let G have one vertex per component of either surface and one edge per glued pair, including isolated vertices. For the quotient W ,*

$$c(W) = c(G), \quad \chi(W) = \chi(S) + \chi(T) - r, \quad (1)$$

$$\delta(W) = \delta(S) + \delta(T) + \beta_1(G), \quad \beta_1(G) = r - c(S) - c(T) + c(G). \quad (2)$$

Hence W is a disk union exactly when S, T are disk unions and G is a forest; it is one disk exactly when the pieces are disk unions and G is a tree.

Proof. Paths between original components cross glued intervals, so quotient connectivity equals graph connectivity. Triangulate compatibly with the intervals. Each glued interval has Euler characteristic one; their disjointness excludes higher-intersection corrections. Inclusion-exclusion gives (1). Substitute into $c(W) - \chi(W)$ to get (2).

Every original boundary circle has points outside its finitely many disjoint proper closed attachment intervals. These survive on the quotient boundary. All quotient components therefore have nonempty boundary. Every term on the right of (2) is nonnegative, and their sum is zero exactly in the stated forest case. The connected case is the tree case. $\square$

This is the topological mechanism used in the predecessor reports [3, 4], recalled to make this strengthening self-contained. No orientation signs are required for the disk case: a leaf disk in a tree can always be oriented compatibly with its neighbor. Positive defect cannot be

repaired by later permitted attachments. A nonterminal component with no live arc cannot join a required nonempty remainder and must be rejected.

2.2 Partitions and certified envelopes

Let U be a set of $r \geq 1$ labels. For $p, q \in \text{Part}(U)$, write $|p|$ for the number of blocks, $p \preceq \sigma$ for refinement, and $p \vee q$ for the least common coarsening. Its blocks are generated by alternating paths within blocks of p and q . A disk-union frontier meeting every live arc determines p : two arcs have the same block exactly when they lie on the same disk.

The bipartite multigraph $G(p, q)$ has a vertex for every block of p or q , and an edge $u \in U$ between its containing blocks. Every vertex meets an edge. Thus

$$D(p, q) = [p \vee q = \mathbf{1}_U \text{ and } |p| + |q| = r + 1] \quad (3)$$

is exactly the tree indicator, and hence the disk-completion indicator under the arc contract by theorem 2.1.

Definition 2.2 (Two-sided connectivity envelope). An envelope is a pair (σ, ρ) of partitions of U . It is safe for a candidate family and continuation language when every candidate has $p \preceq \sigma$ and every permitted complementary disk union has $q \preceq \rho$. Its incidence graph is $H = G(\sigma, \rho)$. The restricted matrix $\mathcal{D}_{\sigma, \rho}$ has rows all $p \preceq \sigma$, columns all $q \preceq \rho$, and entries $D(p, q)$.

A safe envelope can include combinations not simultaneously realizable. It is an overapproximation. Omitting a possible connectivity relation, in contrast, can make pruning unsound. Certifying an envelope is separate from checking a row identity.

Proposition 2.3 (Disconnected-envelope rejection). *If $\sigma \vee \rho \neq \mathbf{1}_U$, then $\mathcal{D}_{\sigma, \rho}$ is zero.*

Proof. For refinements, $p \vee q \preceq \sigma \vee \rho$. The former cannot have one block when the latter has more than one. $\square$

Henceforth suppose H is connected and put

$$s = |\sigma|, \quad t = |\rho|, \quad \lambda = \beta_1(H) = r + 1 - s - t \geq 0. \quad (4)$$

Parallel-edge multiplicity matters: a parallel arc edges between two vertices contribute $a - 1$ independent cycles, not zero.

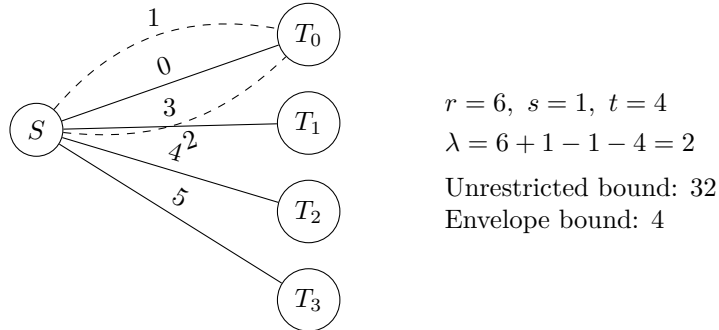


Figure 1: Two independent cycles. Solid edges form a spanning tree and dashed edges are chords. Adding pendant arc edges increases r without increasing λ . All edges remain genuine arcs; none is silently bundled.

3 Cycle-space factorization and exact ranks

3.1 Vertex splitting as a system of cycle constraints

The cycle space $Z(H) \subseteq \mathbb{F}_2^U$ consists of edge vectors with even incidence at every vertex. It has dimension λ . Fix a basis $z_1, \dots, z_\lambda$ and identify $x \in \mathbb{F}_2^\lambda$ with $z(x) = \sum_i x_i z_i$. A refinement $p \preceq \sigma$ splits each σ -vertex into its p -subblocks. Choose one omitted subblock at every coarse vertex. For every other subblock C , impose

$$\ell_C(x) = \sum_{u \in C} z(x)_u, \quad A_p(C, i) = \sum_{u \in C} z_i(u) \in \mathbb{F}_2. \quad (5)$$

The matrix A_p has $j(p) = |p| - s$ rows and λ columns. Define B_q similarly on the ρ -side, with $k(q) = |q| - t$ rows.

Lemma 3.1 (Split cycle-space identity). *Under the common edge-label identification,*

$$Z(G(p, q)) = \left\{ z(x) : \begin{pmatrix} A_p \\ B_q \end{pmatrix} x = 0 \right\}, \quad \beta_1(G(p, q)) = \lambda - \text{rank}_{\mathbb{F}_2} \begin{pmatrix} A_p \\ B_q \end{pmatrix}. \quad (6)$$

Proof. Even incidence at every split vertex implies even incidence at the original vertex by summing its subblock equations. Hence every split-graph cycle vector lies in $Z(H)$. Conversely, a vector in $Z(H)$ satisfies the parity equation at each original vertex. The sum of all its subblock equations is therefore zero. Imposing all but the omitted equation forces the omitted one too. This holds at each left and right vertex. The remaining equations are exactly the displayed stacked system. The cycle-basis coordinate map is an isomorphism, so rank-nullity gives the dimension identity. $\square$

No independence of the remaining equations is assumed. Splitting can disconnect the graph; then the constraints can be dependent and determinant-zero rows arise naturally.

Lemma 3.2 (Determinantal disk predicate). *If $j(p) + k(q) = \lambda$, then*

$$D(p, q) = \det_{\mathbb{F}_2} \begin{pmatrix} A_p \\ B_q \end{pmatrix}. \quad (7)$$

If $j(p) + k(q) \neq \lambda$, then $D(p, q) = 0$.

Proof. The split graph has r edges and $s + t + j(p) + k(q)$ vertices. The tree edge-vertex condition is equivalent to $j(p) + k(q) = r + 1 - s - t = \lambda$. If it fails, the disk predicate is zero. If it holds, the graph has $r + 1$ vertices. A graph with r edges and $r + 1$ vertices is a tree exactly when its cycle space is zero: a forest with these counts has one component. By theorem 3.1, this is equivalent to nonsingularity of the square stacked matrix. Over $\mathbb{F}_2$ the determinant is zero or one. The empty 0×0 determinant is one, which includes $\lambda = 0$. $\square$

3.2 Explicit exterior coordinates

Let $[\lambda] = \{1, \dots, \lambda\}$, with $[0] = \emptyset$. For every $J \subseteq [\lambda]$, define

$$F_\sigma(p, J) = \begin{cases} \det A_p[:, J], & |J| = j(p), \\ 0, & |J| \neq j(p). \end{cases} \quad (8)$$

If $j(p) > \lambda$, the whole row is zero. Define F_ρ using B_q , and let $\mathcal{J}$ be the permutation matrix of $J \mapsto [\lambda] \setminus J$.

Theorem 3.3 (Cycle-envelope factorization). *Over $\mathbb{F}_2$,*

$$\mathcal{D}_{\sigma,\rho} = F_\sigma \mathcal{J} F_\rho^\top. \quad (9)$$

The feature dimension is 2^λ , and rows of grade $j(p)$ are supported in a coordinate space of dimension $\binom{\lambda}{j(p)}$.

Proof. For noncomplementary grades, no subset has size $j(p)$ while its complement has size $k(q)$, so the pairing is zero. For complementary grades, Laplace expansion of the stacked determinant by its first $j(p)$ rows gives

$$\det \begin{pmatrix} A_p \\ B_q \end{pmatrix} = \sum_{\substack{J \subseteq [\lambda] \\ |J|=j(p)}} \det A_p[:, J] \det B_q[:, [\lambda] \setminus J].$$

All signs disappear in characteristic two. Apply theorem 3.2. Homogeneous support in (8) gives the graded dimension. $\square$

Proposition 3.4 (Exactly which rows vanish). *For $p \preceq \sigma$, put $a = c(G(p, \rho)) = |p \vee \rho|$. Then*

$$\text{rank } A_p = j(p) - a + 1. \quad (10)$$

Consequently $F_\sigma(p, \cdot) \neq 0$ exactly when $p \vee \rho = \mathbf{1}_U$.

Proof. Use theorem 3.1 with $q = \rho$, so there are no right-side constraints. The cycle rank of that graph is $r - (|p| + t) + a = \lambda - j(p) + a - 1$. Subtracting from λ proves (10). A matrix with $j(p)$ rows has a nonzero maximal minor exactly when it has row rank $j(p)$, including the empty-row case. Thus a nonzero feature is equivalent to $a = 1$. $\square$

This provides a cheap connectivity prefilter before feature construction. The delivered unrestricted-root and exact-state comparators receive this same prefilter; the measured improvement is not obtained by withholding an obvious rejection rule.

3.3 Sharpness for every connected envelope

Choose a spanning tree T of H , with chords $e_1, \dots, e_\lambda$. Their fundamental cycle basis satisfies $z_i(e_j) = [i = j]$. Every original vertex has a tree edge. For $S \subseteq [\lambda]$, construct $p_S \preceq \sigma$ by detaching the left endpoint of each chord indexed by S into a singleton subblock. All other incident labels stay in a core subblock, which is nonempty because it retains a tree edge. Similarly define $q_R \preceq \rho$ by detaching right endpoints indexed by R .

Lemma 3.5 (Chord-detachment witnesses). *With core subblocks omitted in the feature construction,*

$$F_\sigma(p_S, \cdot) = e_S, \quad F_\rho(q_R, \cdot) = e_R.$$

Independently of coordinate choices,

$$D(p_S, q_R) = [R = [\lambda] \setminus S]. \quad (11)$$

Proof. A detached singleton chord imposes the coordinate equation $x_i = 0$, giving the stated unit exterior vector. For a direct graph proof, a chord detached at both ends becomes a separate component while the core spanning tree remains. If the detachment sets are disjoint but do not cover all chords, their total size is less than λ , so the tree edge–vertex count fails. If they partition the chords, the core is T and each former chord is now a pendant edge attached at its undetached endpoint. The result is a tree. $\square$

Theorem 3.6 (Exact total and graded ranks). *For every connected envelope,*

$$\text{rank}_{\mathbb{F}_2} \mathcal{D}_{\sigma, \rho} = 2^\lambda. \quad (12)$$

The block with $|p| = s + j$ and $|q| = t + \lambda - j$ has exact rank $\binom{\lambda}{j}$ for $0 \leq j \leq \lambda$.

Proof. The factorization gives both upper bounds. The rows p_S and columns q_R form a subset-complement permutation matrix of order 2^λ . Restricting to $|S| = j$ and $R = [\lambda] \setminus S$ gives a permutation submatrix of order $\binom{\lambda}{j}$ in that grading. These lower bounds match the upper bounds. When $\lambda = 0$, the single distinguished entry is $D(\sigma, \rho) = 1$. $\square$

This is a binary-rank statement, not a claim about all field characteristics. The sharp families are realizable as abstract polygonal disk unions with separated labelled arcs. Simultaneous realization in one embedded knot-region type remains a geometric question.

3.4 Cycle rank as a representable overlap parameter

Proposition 3.7 (Linear-matroid interpretation). *For a partition τ , let*

$$V_\tau = \text{span}_{\mathbb{F}_2} \{e_u + e_v : u, v \text{ belong to one block of } \tau\} \subseteq \mathbb{F}_2^U.$$

Then $\dim V_\tau = r - |\tau|$, and for a connected envelope,

$$Z(H) = V_\sigma \cap V_\rho, \quad \dim(V_\sigma \cap V_\rho) = \lambda. \quad (13)$$

Proof. In a block C , the differences from one fixed element to the others form a basis of vectors with even coordinate sum on C . Hence V_τ is exactly the space of vectors having even sum on every τ -block, of dimension $\sum_C (|C| - 1) = r - |\tau|$. The simultaneous parity conditions for σ and ρ are exactly those of $Z(H)$. Alternatively, $V_\sigma + V_\rho = V_{\sigma \vee \rho}$ because differences propagate along join paths. The connected join has dimension $r - 1$, and the intersection dimension formula gives $r + 1 - s - t$. $\square$

Taking separately colored spanning forests inside the σ - and ρ -blocks gives a graphic-matroid interpretation: their rank sum minus the rank of their union is λ . This relates the parameter to established connectivity and representative-set methods [9, 7]; the factorization here makes its disk-specific behavior explicit.

4 Weighted representatives and independently checked pruning

4.1 The optimization guarantee

A candidate a consists of an actual partial-surface witness, its partition p_a , and an integer cost w_a written in binary. All candidates in one reduction call have a fixed geometric type. For a finite family $\mathcal{A}$ and $q \preceq \rho$, put

$$\text{opt}(\mathcal{A}, q) = \min\{w_a : a \in \mathcal{A}, D(p_a, q) = 1\}, \quad \min \emptyset = +\infty.$$

A subset represents $\mathcal{A}$ when it preserves this optimum for every such q . Costs may be negative, since the candidate family and assembly language are finite.

Theorem 4.1 (Optimal cycle-envelope representative bound). *If every $p_a \preceq \sigma$ and the envelope is connected, a deterministic algorithm retains a representative subset $\mathcal{A}'$ of original candidates satisfying*

$$|\mathcal{A}'| \leq 2^\lambda, \quad |\{a \in \mathcal{A}' : |p_a| = s + j\}| \leq \binom{\lambda}{j}. \quad (14)$$

It scans in nondecreasing cost and retains a candidate exactly when its feature row is independent of previously retained rows. Zero rows are discarded.

Proof. Fix a total order refining cost order. Gaussian elimination retains an independent family; homogeneous support in theorem 3.3 gives the size bounds. Every discarded row has an expansion

$$F_\sigma(p_a, \cdot) = \sum_{b \in E_a} F_\sigma(p_b, \cdot), \quad b \in \mathcal{A}', \quad w_b \leq w_a. \quad (15)$$

For any cap completing a , pair this identity with $\mathcal{J}F_\rho(q, \cdot)^\top$. The left side is one, so at least one summand is one. A retained candidate of no greater cost therefore completes that same cap. Apply this to an optimum candidate. The reverse inequality holds because the retained family is a subset. Infeasibility is preserved for the same reason. $\square$

The parity calculation finds a replacement for each particular successful row; it is not a randomized parity-of-solutions decision. A retained row refers to an original witness, not an XOR of surfaces.

Theorem 4.2 (Sharpness for all representative-subset methods). *For every connected envelope there is a cost-zero family requiring all 2^λ representatives to preserve every allowed cap query. In grade j there is such a family requiring all $\binom{\lambda}{j}$ candidates.*

Proof. Take one candidate for every p_S in theorem 3.5. The cap $q_{[\lambda] \setminus S}$ completes exactly that candidate among the family. Omitting it changes its query from feasible to infeasible. Restrict to $|S| = j$ for the graded statement. $\square$

This is not a running-time lower bound for arbitrary algorithms. One explicitly supplied cap can be checked against an explicit candidate list by a direct polynomial-time scan; no exponential-dimensional preprocessing is necessary for that isolated task.

4.2 Sectors, coordinate changes, and certificates

Suppose an auxiliary label lies in Q sectors and the continuation condition determines the required sector independently of internal partition. Reduction separately in each sector preserves the corresponding queries and uses at most $Q2^\lambda$ candidates. The implementation uses two-bit sectors with XOR composition, so $Q \leq 4$. They are actual boundary-homology sectors only under the geometric chain contract in section 7.

A pruning certificate lists retained original indices and an expansion (15) for every source row. It binds the ordered source partitions, costs, sectors, witnesses, and envelopes. The checker reconstructs the feature rows, verifies each expansion, checks cost and sector constraints, and checks the graded size bounds. Independence helps produce a small family, but these expansion identities and cost inequalities suffice to certify representativity.

Proposition 4.3 (Coordinate independence of pruning identities). *Changing the cycle-space basis or the omitted subblock at any coarse vertex leaves the validity of homogeneous linear expansion identities unchanged.*

Proof. A cycle-basis change right-multiplies every constraint matrix by one fixed invertible matrix. Its action on degree- j maximal minors is the invertible map on the j th exterior power. Linear identities in that grade are therefore preserved.

At a coarse vertex, the sum of all subblock parity rows is zero. Changing the omitted subblock changes its $m - 1$ listed rows by an invertible linear combination. The determinant of that transformation over $\mathbb{F}_2$ is one. Row permutations also have determinant one. Thus the wedge of all constraint rows, and all its maximal minors, is unchanged, even when the evaluated rows are dependent. Different grades have disjoint coordinate supports, so identities decompose by grade. $\square$

The producer computes fundamental cycles and sparse exterior products. The checker instead obtains a cycle basis by row-reduced nullspace elimination and computes features by enumerating square minors. It omits the last subblock instead of the first. Theorem 4.3 justifies these genuinely different coordinate choices.

A SHA-256 digest is convenient source binding, not a proof by itself. The checker receives the actual ordered source family independently and reconstructs the payload. A valid row certificate still does not prove that the supplied envelopes cover every geometric continuation. That is a separate compiler or producer assertion.

4.3 Encoded local cost

Let L be the supplied candidate count, B the maximum cost bit length, and W their explicit witness encoding length. Put $d = 2^\lambda$. A spanning tree and fundamental cycles are constructed in polynomial time in r, λ . A feature can be computed by sparse exterior multiplication in $\text{poly}(r, \lambda)d$ bit work. Alternatively, enumerate its $\binom{\lambda}{j}$ square minors using binary elimination, giving the same single-exponential form with a different polynomial.

Naive elimination uses at most d pivots per row and XORs d -bit vectors. Tracking expansions into retained rows costs the same order. Thus a conservative local bound is

$$\text{poly}(r, \lambda) + O\left(Ld^2 + L \text{poly}(r, \lambda)d + LB \log(L + 1) + W\right), \quad (16)$$

with additional polynomial charges for source encoding. The independent checker can use the same Ld^2 expansion bound and $L \text{poly}(r, \lambda)d$ minor-construction bound.

The basis and its binary expansions occupy $O(d^2)$ bits. An explicit certificate with at most d indices per input row uses $O(Ld \log(L + 1))$ bits. Source candidates and witnesses remain additional storage. A d -bit packed-integer XOR is charged $O(d)$ bit operations, not treated as constant time. There is no pseudo-polynomial dependence on the numerical values of the costs.

An explicitly Bell-sized input family must still be read. The next section turns this family-level reduction into a search theorem by generating later candidates only from already reduced families and an explicit finite option set.

5 A complete finite grammar and polynomial envelope compiler

5.1 The supplied language

Fix nonempty interfaces $U_0, \dots, U_h$ with $r_i = |U_i|$. A layered grammar supplies an explicit initial family of disk unions on U_0 ; for each $i = 1, \dots, h$, an explicit list of disk-union patches on $U_{i-1} \sqcup U_i$; and an explicit family of disk-union caps on U_h . Choose one initial object, one

option at every layer, and one cap. Costs add and sectors XOR. Matching incoming and outgoing arc labels are glued. Terminal success is one disk with an allowed final sector.

Every local option is available independently of the current partition. Topological checks reject attachment cycles or sealed components, but there are no hidden same-component guards. For an ambient application, sufficient geometric types must ensure that the same outside objects and arc maps are legal for every candidate of the type. Boundary or embedding information visible to the continuation belongs in that type.

The reference solver is complete for this explicit abstract language, not automatically for disks in a knot exterior. Its result names are deliberately

FOUND_ABSTRACT_DISK NO_DISK_IN_GRAMMAR.

Neither is automatically an unknot or knottedness verdict.

5.2 Forward and backward envelope recurrences

For disjoint sets, $\alpha \oplus \text{disc}(V)$ denotes the partition equal to α on one set and discrete on V . Restriction followed by canonical relabelling is denoted $\tau|_V$. Write $\mathcal{P}_i$ for the layer- i partitions and $\mathcal{C}$ for the cap list. Define

$$\sigma_0 = \bigvee_{a \in \mathcal{A}_0} p_a, \quad (17)$$

$$\sigma_i = \left((\sigma_{i-1} \oplus \text{disc}(U_i)) \vee \bigvee_{\pi \in \mathcal{P}_i} \pi \right) |_{U_i}, \quad 1 \leq i \leq h, \quad (18)$$

$$\rho_h = \bigvee_{q \in \mathcal{C}} q, \quad (19)$$

$$\rho_{i-1} = \left((\text{disc}(U_{i-1}) \oplus \rho_i) \vee \bigvee_{\pi \in \mathcal{P}_i} \pi \right) |_{U_{i-1}}, \quad h \geq i \geq 1. \quad (20)$$

The join of an empty option list is discrete on its specified label set. Empty languages are allowed and yield no success.

Theorem 5.1 (Constructive safe envelopes). *Every realizable prefix partition at U_i refines σ_i , and every realizable complementary suffix partition meeting U_i refines ρ_i . Both sequences are computable in polynomial time in the explicit grammar length, without enumerating prefix or suffix assemblies.*

Proof. The initial statement follows from (17). Suppose a prefix partition p refines σ_{i-1} . For a chosen patch π , quotient connectivity on outgoing arcs is the restriction of $(p \oplus \text{disc}(U_i)) \vee \pi$. Join and restriction preserve refinement. Replacing p by σ_{i-1} and π by the join of every option only coarsens that partition. This proves the forward induction. The suffix induction is identical in reverse order, starting from the cap family.

Each join connects labels in every explicitly listed block. Disjoint-set union or graph connectivity computes it in polynomial time in the listed incidences. No Cartesian-product candidate family is generated. Repeating over all layers is polynomial in input length. $\square$

The compiler joins mutually exclusive options. This may create impossible connections and overestimate λ_i , losing compression but not correctness. Computing the smallest envelope of all realizable successful contexts is a different problem and is not silently assumed solved.

The independent checker uses a universal graph on all interface labels, not the recursive disjoint-set compiler. It separately takes whole-prefix and whole-suffix connectivity and restricts to each cut. Both constructions express the transitive closure of the same local block relations; their equality is also checked on randomized grammars.

5.3 Transitions and context-safe replacement

To attach a current disk union to a patch, form their component-incidence graph with one edge per incoming arc. Reject an attachment cycle. Otherwise its graph components are disks by theorem 2.1. Reject any component with no outgoing arc, since nonempty later material must still join the final disk. Restrict surviving component labels to outgoing arcs to obtain the next partition. A patch component having only outgoing arcs is allowed: it is a new disk, initially an isolated vertex in this attachment graph.

Identical partition/sector states can first be deduplicated by keeping the cheapest witness. Every solver mode uses the same deterministic deduplication, transition generator, and safe filter $p \vee \rho_i = \mathbf{1}_{U_i}$. Cycle mode then applies the representative theorem to (σ_i, ρ_i) .

Theorem 5.2 (Context-safe cycle-envelope replacement). *Replacing an interface family by representatives for its safe two-sided envelope preserves the minimum cost of every successful terminal assembly. Repeated replacements throughout the supplied grammar preserve global minimum cost and feasibility.*

Proof. Fix a successful complete assembly using a candidate S at the replaced interface. Glue all pieces outside S according to the selected completion to obtain T . Its entire contact with S is the labelled interface of separated arcs. Since the final surface is a disk, theorem 2.1 makes both S, T disk unions with a tree attachment graph. Every component of T meets the interface, or the final assembly would be disconnected. Its cap partition q therefore exists, and the compiler theorem gives $q \preceq \rho_i$.

Representativity supplies an actual retained S' of the same type and sector, of no greater cost, completed by that same q . All remaining options and arc maps stay legal because there is no unrecorded partition-dependent guard; in an ambient application this is the geometric type-substitution contract. The union $S' \cup T$ is a disk. Its intermediate surfaces cannot have positive defect, since such defect cannot later disappear. Nor can they have a sealed component followed by nonempty material. Thus the replacement path survives every local transition and liveness test.

Cost additivity preserves or improves total cost. Conversely a retained subset cannot introduce an assembly not previously present. Apply this argument successively in the finite evaluation order to obtain equality of global optima. $\square$

This proof concerns whole remaining contexts, not identical behavior under each individual next operation. Different caps may use different retained replacements for one discarded candidate. That is why representatives can be much fewer than physical-state equivalence classes.

5.4 The complete run certificate

A run certificate binds the grammar and supplies one pruning certificate at each interface, the final optimum, original option indices, and work counters. The checker reconstructs source candidate lists itself: it tries every supplied option on every previously retained candidate using an independent breadth-first component-graph transition, performs the specified cheapest-state

deduplication, and checks the reduction certificate against that reconstructed list. It does not trust a producer's alleged list of generated candidates. At the end it evaluates every cap query directly.

This validates positive and negative answers *inside the finite grammar*. Omitting an option from the certificate cannot justify an empty table, because options are independently read from the source grammar. It does not prove that a geometric producer listed every patch needed in a knot exterior. That completeness obligation lies outside the language being checked.

Independent PL replay then expands the selected disks into triangulated polygons, glues marked boundary edges, checks edge incidences and vertex links, and computes components, Euler characteristic, boundary circles, and orientability. This is more than repeating the partition predicate, but still proves abstract topology rather than ambient embedding.

6 Bit complexity, large interfaces, and adaptive restrictions

6.1 End-to-end accounting for the supplied grammar

Let $I \geq 2$ be the explicit structural bit length, including all partition lists, interfaces, option indices, sectors, and source data. Let $B \geq 1$ bound input cost bit lengths. Accumulated costs have $O(B + \log(h + 2))$ bits. Let K_i be the number of layer- i options and L_0 the initial candidate count. These structural quantities and total raw arc count are charged to I .

A disconnected envelope certifies no disk in the supplied grammar by theorem 2.3. Otherwise put

$$\lambda_i = r_i + 1 - |\sigma_i| - |\rho_i|, \quad \lambda_* = \max_{0 \leq i \leq h} \lambda_i.$$

Let $Q \leq 4$ be the sector count. More general finite type sets are additional explicit parameters, not hidden constants.

Theorem 6.1 (Cycle-envelope parameterized search). *The supplied layered grammar can be optimized deterministically, with source-bound pruning certificates and recovered witnesses, in*

$$\text{poly}(I, B)2^{3\lambda_*} \quad (21)$$

bit time. The same form bounds checking the run certificate and expanded witness. The reference implementation satisfies this bound up to polynomial factors when its allocation guards are disabled or not reached.

Proof. Compile envelopes in polynomial time. Constructing all cycle bases costs polynomial work in the explicit arc counts. Initial reduction costs $\text{poly}(I, B)4^{\lambda_0}$, charging $L_0 \leq I$, by (16).

At layer i , at most $Q2^{\lambda_{i-1}}$ candidates enter. Trying every K_i option produces at most $K_i Q 2^{\lambda_{i-1}}$ transitions. Connectivity, liveness, canonicalization, arithmetic, and explicit witness-path copying cost polynomial work in I, B per transition. Output reduction costs at most

$$\text{poly}(I, B)K_i Q 2^{\lambda_{i-1} + 2\lambda_i}.$$

Deduplication and sorting add polynomial factors: the logarithm of the generated family size is $O(\log I + \lambda_*)$, and $\lambda_* \leq I$. Summing gives the more detailed bound

$$\text{poly}(I, B) \left(L_0 4^{\lambda_0} + \sum_{i=1}^h K_i Q 2^{\lambda_{i-1} + 2\lambda_i} + Q|C|2^{\lambda_h} + 1 \right), \quad (22)$$

which implies (21). Displaying structural quantities outside the polynomial here exposes the nonuniform exponent profile; initial-list reading is still charged.

The checker reconstructs the same number of transitions. Its independent minor and expansion calculations satisfy the same single-exponential local bound. Witnesses select one explicitly listed patch at each layer, so their polygonal expansion has polynomial size in I . Vertex-link and surface computations are polynomial in that expanded size. Correctness follows from context-safe replacement and the final direct cap check. $\square$

This deliberately elementary bound uses neither fast matrix multiplication nor optimized representative-product algorithms. A supplied binary assembly grammar also admits a $2^{O(\lambda^*)}$ argument by combining two reduced child tables, provided every node has certified envelopes for its whole context and explicitly bounded type/rule costs. The executable compiler and full run checker here implement only the layered case.

Keeping all stage certificates costs at most $\text{poly}(I, B)4^{\lambda^*}$ bits under this accounting, in addition to explicit source storage. Streaming could reduce history but is not the delivered storage policy. A binary-compressed syntax describing exponentially many unexpanded options is not an explicit polynomial-size grammar by fiat.

6.2 Unbounded raw width at fixed cycle rank

Corollary 6.2 (Tree appendages cost no exponential dimension). *Appending arbitrarily many pendant edges with new endpoints to a connected envelope, or subdividing an edge by an odd-length path to preserve the bipartite sides, leaves λ unchanged. The exact restricted rank stays 2^λ , while the genuine arc count can grow arbitrarily.*

Proof. A pendant edge adds one vertex and one edge and preserves connectivity. Subdivision adds equal numbers of edges and vertices. Thus $|E| - |V| + 1$ is unchanged. Any bipartite graph without isolated vertices is the incidence graph of the partitions of its edge labels by left and right endpoints. Apply the exact-rank theorem to these partitions. $\square$

For example, use one left vertex, $\lambda + 1$ parallel edges to one right vertex, and $r - \lambda - 1$ additional pendant right vertices. Then

$$\sigma = (0, \dots, 0), \quad \rho = (\underbrace{0, \dots, 0}_{\lambda+1}, 1, 2, \dots, r - \lambda - 1).$$

The exact rank is 2^λ for every $r \geq \lambda + 1$. The wide-interface test uses $r = 256$, $\lambda = 3$, and eight sharp chord-detachment candidates. Producer and independent checker operate on eight-dimensional features, not 2^{255} -dimensional features. All 256 arcs remain explicitly represented and read.

6.3 Narrowing is safe; widening is not

Proposition 6.3 (Monotone restriction of the context language). *A family representative for all caps refining ρ remains representative when that allowed cap set is restricted. If safe envelopes are replaced by refinements $\sigma' \preceq \sigma$, $\rho' \preceq \rho$ and the new incidence graph stays connected, then*

$$\lambda' = r + 1 - |\sigma'| - |\rho'| \leq \lambda.$$

A new reduction can use this no-larger dimension on candidates still refining the new past envelope. Widening the future language has no analogous guarantee.

Proof. Equality of optimum functions on a set of queries implies equality on a subset. Refinement increases block counts, proving the displayed inequality; apply the representative theorem to the new envelope. The following example proves the last claim. $\square$

For two arcs, take $\sigma = 00$, old $\rho = 01$, hence $\lambda = 0$. Candidates 00 of cost zero and 01 of cost one reduce to only 00 . The newly allowed cap 00 completes 01 but not 00 . An adaptive algorithm widening future options must return to an unpruned source or have retained representatives for an envelope already containing the larger language. A resource limit cannot establish that omitted future options are impossible.

Restricting *past* candidates deserves a separate warning. One must not simply filter an old representative subset by a new past predicate: representatives preserve cap queries, not arbitrary filters on source candidates. Rebuild from the filtered source family, or prove that its needed representatives survive. Refining an envelope around the family actually being reduced is safe; forgetting already discarded source witnesses under a new eligibility condition is a different operation.

7 The link to quasi-polynomial unknot recognition

7.1 A conditional theorem with explicit geometric obligations

The local and finite-grammar theorems are unconditional. Turning them into a recognizer requires a producer connecting the language to the input knot. The following statement exposes those remaining obligations.

Corollary 7.1 (Low-cycle-envelope recognition route). *Suppose every validated n -crossing knot diagram admits a grammar of the preceding kind, constructible in polynomial bit time, with polynomial explicit size, local geometric validation cost, input-cost bit length, and expanded witness-verification cost. Its rules are sound for embedded arc attachments in the source-certified exterior and it generates an essential-boundary properly embedded disk if and only if one exists there. Assume geometric types satisfy the context-substitution contract, and all type and rule multiplicities are polynomially bounded and charged. If the safe envelopes used by the search have*

$$\lambda_* = O((\log(n+2))^2), \tag{23}$$

then the resulting unknot recognizer runs in $n^{O(\log n)}$ bit time for $n \geq 2$. If $\lambda_ = O(\log(n+2))$, the same route is polynomial time.*

Proof. Apply theorem 6.1, charging the stipulated geometric costs. Polynomially many fixed types multiply its bounds polynomially: reduce each type separately and try all explicitly legal typed transitions. Safe envelopes can be compiled after forgetting finite type restrictions, since this only adds possible connections; tighter type-specific envelopes require their own safety proof. The same context argument applies within a type. The executable implementation handles the fixed layer type and the four XOR sectors; this finite-control extension does not assert additional implemented functionality.

Under (23), $2^{3\lambda_*} = \exp(O(\log^2 n)) = n^{O(\log n)}$; the smaller condition gives a polynomial factor. A knot exterior in S^3 has compressible boundary exactly for the unknot [10]. The assumed source soundness and disk completeness therefore turn the grammar decision into the knot decision. $\square$

The change from the predecessor sufficient condition is substantive: raw arc counts may be polynomial rather than polylogarithmic, provided two-sided cycle overlap is small. Theorem 6.2

makes this a strictly weaker abstract restriction, not just different notation for raw width. In the worst case $\sigma = \rho = \mathbf{1}_U$ and $\lambda = r - 1$, so a universal improvement still needs a structural theorem for the produced envelopes.

Neither that producer nor (23) for all knot diagrams is proved here. The existing source-checked canonical exterior should be reused. The unproved integration is a complete, type-controlled disk-assembly search with a proved cycle-overlap bound, not another assertion that an exterior can be triangulated.

7.2 Essential boundary and two-bit sectors

Let the source exterior have boundary torus $\mathbb{T}$. Suppose a certified basis of $H^1(\mathbb{T}; \mathbb{F}_2)$ is provided and every patch carries evaluations of its external boundary chain, with correct seam cancellations and any necessary local correction terms. These two evaluations compose by XOR. Only under this chain-level contract do the solver’s generic two-bit sectors become genuine homological sectors.

Proposition 7.2 (Essential-boundary refinement). *Separate reduction in the four boundary parity sectors preserves the minimum-cost disk completion in every prescribed final sector. A terminal properly embedded disk in the source exterior has essential boundary exactly when its sector is nonzero.*

Proof. For a fixed complement, its sector determines the partial sector required for each prescribed final value. Representativity within that sector preserves the optimum and additivity preserves the total label.

The boundary of a properly embedded disk is an embedded circle on $\mathbb{T}$. An inessential circle is null-homologous. An essential embedded torus circle has primitive integral class (a, b) , so its coordinates cannot both be even. Its reduction modulo two is nonzero. Conversely nonzero mod-two homology rules out an inessential circle. $\square$

The implementation does not construct the torus cocycles or export an embedded normal disk. Two stored bits named “homology” do not establish the needed chain contract. Aggregate parity of a multi-component normal surface can also cancel; it does not replace the native componentwise disk criterion described in the previous work [3].

7.3 Integration gate rather than automatic runtime patch

Native integration should first export actual disk pieces and separated arc interfaces from a source-certified region decomposition. It must expose all geometric types and prove that each type admits the same outside attachment choices. The explicit complete rule source should feed the envelope compiler and run checker. Finally a recovered abstract witness must replay as an embedded disk with nonzero certified boundary sector in the *same* input exterior.

No remote repository changes were made. A runnable native checkout was not available in the execution environment, so no maintained recognizer regression suite or native timing corpus was run. This is an additive research package with proposed integration notes, not a claim that production dispatch already uses the kernel.

For an incomplete geometric grammar, not finding a disk is inconclusive about knottedness. A successful abstract gluing is not an unknot certificate without embedding and source binding. Resource exhaustion must remain inconclusive and preserve access to the maintained exact fallback.

8 Safety boundaries and counterexamples

8.1 Arc multiplicity and operations outside the model

Two disks glued along two separated arcs have Euler characteristic zero. Suitable seam maps produce an annulus or a Möbius band; neither is a disk. Treating the two arcs as one predicts Euler characteristic one incorrectly. The cycle-envelope parameter does not permit this compression. It counts cycles on genuine arcs and removes exponential, not polynomial, dependence on tree-like arc structure.

Whole-boundary-circle gluing, capping a circle, compression surgery, and point quotients do not satisfy this same arc-defect identity. Capping can remove a deficit in ways excluded by monotonicity. Such operations require a new theorem, not a change of names.

8.2 Hidden guards and expensive geometric types

At width three with unrestricted envelopes, the feature rows of 001,010,011 have a linear dependence: the third is the sum of the first two in its grade. With costs 0, 0, 1, valid reduction may discard 011. But a rule enabled only when labels 1 and 2 lie on the same component then loses its only admissible source. Preserving every disk-cap optimum does not preserve every partition-dependent guard. The context contract is necessary; the same example warns against blindly filtering a previously reduced family.

Recording the guarded relation in the type can restore correctness, but all extra types must be counted. Cyclic order, embedding position, or local normal quadrilateral choices can cause similar growth. A small 2^λ factor within each of exponentially many uncharged types does not give a quasi-polynomial algorithm.

8.3 Scalar optimization is not counting or full state equivalence

Representative subsets preserve scalar minima and existence for the specified caps. They do not preserve the number of successful assemblies or all Pareto-optimal values of several costs. Ties can produce different optimum witnesses; the theorem preserves cost, not lexicographic preference. Tests compare minimum cost and feasibility and replay whatever original witness each solver actually returns.

An envelope also need not be the smallest observation space for the *actual* suffix language. A single supplied cap gives a matrix with at most one column, even if all refinements of its envelope would have rank 2^λ . The exact theorem is sharp for the full refinement-closed cap language, not every subset. This leaves room for stronger suffix-specific methods and explains the direct-scan control below.

Source hashes and pruning identities do not prove ambient embedding, normal matching, or producer completeness. Literal surface replay does not establish where the surface embeds. These distinctions remain after all finite checks pass.

9 Implementation and executed evidence

9.1 Test coverage and independent audits

The source uses only the Python standard library and syntax compatible with Python 3.10 or later. The executed environment was CPython 3.13.5 on Linux x86_64 with glibc 2.41. The

retained suite has **42 passing test methods**; loop iterations are not counted as additional methods.

Tests cover every envelope pair through width four, direct and graded matrix ranks, chord-detachment witnesses, exact zero-row characterization, alternative cycle bases, 4096-bit signed costs, source and certificate mutations, sectors, resource exceptions, wide low-cycle interfaces, independent envelope compilation, small exhaustive transitions, complete grammar runs, and literal polygonal surface checks. Negative controls include future-language widening, annuli, twisted bands, reused arcs, wrong source option lists, and forged terminal costs or outcomes.

The larger fixed-seed audit gives the following counts.

Independent audit item	Completed checks
Envelope matrices	3,162
Direct matrix entries	633,531
Graded rank checks	3,556
Weighted families	1,000
Weighted cap/sector queries	16,904
Complete grammar triples	600
Successful grammar mesh replays	639
Random surface pairs	1,000
Fully enumerated small grammars	40
Literal complete assignments	1,412

Table 1: Executed counts from `results/audit.json`; every check passed.

The matrix audit covers every envelope pair through $r = 5$, plus every right envelope with $\sigma = \mathbf{1}_U$ at $r = 6$. Ranks are computed from an independent graph predicate, not inferred from the proposed features. There are 1,478 zero matrices and the expected nonzero ranks 1, 2, 4, 8, 16, 32. Weighted queries include negative costs, infeasible queries, and all four sectors.

Each of 600 grammar comparisons runs exact, unrestricted-root, and cycle modes. The 639 successful mesh replays count successful solver outputs across those modes, not 639 distinct source grammars. Another 40 grammars are fully enumerated through literal surface replay of all 1,412 assignments, without trusting the producer’s intermediate transitions. These checks support implementation reliability; they neither replace universal proofs nor constitute proof-assistant verification.

9.2 The independent PL oracle

For a disk block carrying k labels, the oracle constructs a polygon with $4k$ boundary vertices, triangulates it from a new center, and marks every fourth boundary edge as an arc. Unused boundary edges separate the arcs. It identifies seam endpoints, reconstructs quotient triangles and edges, and checks every edge has one or two incident triangles and every vertex link is a connected path or circle. It then computes components, Euler characteristic, boundary circles, and orientation consistency by triangle propagation.

The retained grammar example has 141 quotient vertices, 300 edges, and 160 triangles. It is connected and orientable with one boundary component and Euler characteristic $141 - 300 + 160 = 1$: an abstract disk. The example is not asserted to encode any particular knot exterior.

9.3 Paired complete-grammar measurements

For $r = 6, 7, 8$ with $\lambda = 2$, the benchmark supplies all partitions initially, with deterministic seeded costs. The final cap has one three-label block and all remaining labels singleton. Each of 12 layers offers an identity patch and a merge patch for each pair in that three-label block. The compiled cycle rank is two throughout. Initial input sizes are 203, 877, and 4,140 candidates. Fixture construction is excluded from every timed solver; reading and processing the explicit lists is included. No single-exponential generation claim is made for these Bell-sized inputs.

All modes share the grammar, option order, transitions, connectivity prefilter, arithmetic, deduplication, and witness handling. Root mode uses the strong unrestricted 2^{r-1} representative method, not an unnecessarily weak partition algorithm. Cycle mode changes the feature representation. A broad-envelope control uses $r = 6$, $\lambda = 5$, and four layers, where the two algebraic dimensions coincide.

Each mode is measured five times with alternating forward and reverse run order. The medians include compilation and pruning-certificate production, but exclude separate certificate checking and final mesh replay equally for all modes. Those independent checks were performed after timing and their costs are recorded in the raw JSON.

r	λ	Layers	Exact (ms)	Root (ms)	Cycle (ms)	Root/cycle
6	2	12	29.44	22.71	7.47	3.04
7	2	12	85.07	43.95	19.35	2.27
8	2	12	271.24	114.83	77.24	1.49
6	5	4	111.74	35.04	30.44	1.15

Table 2: Median complete abstract-grammar search times. Root/cycle compares the unrestricted-basis median to the cycle-envelope median.

r	λ	Initial input	Exact peak	Root peak	Cycle peak	Exact/cycle work
6	2	203	52	16	4	7.09
7	2	877	130	22	4	6.45
8	2	4140	340	29	4	4.51
6	5	203	203	32	32	5.74

Table 3: Peak stored candidates and generated-candidate work. The work ratio includes the explicit initial list.

On the restricted fixtures cycle mode keeps four candidates. Root-mode peak tables have 16, 22, and 29, while exact peaks have 52, 130, and 340. Root/cycle median-time ratios are 3.04, 2.27, and 1.49. Larger raw width increases initial-list processing, so a larger table reduction need not give a larger wall-time gain. In the broad control both bases keep 32 candidates; timing differences there are implementation constants, not a better asymptotic bound.

For one supplied cap at $r = 8$, direct scanning has median 10.02 ms; unrestricted preprocessing plus the query takes 56.66 ms, and cycle preprocessing plus the query takes 36.91 ms. This is a negative result for preprocessing without continuation reuse. These small host-specific measurements are not a representative geometric workload census or native knot-recognizer speedups. Raw samples are retained.

9.4 Resource and reproducibility policy

Default allocation guards are $\lambda \leq 18$ for cycle features and $r \leq 20$ for the unrestricted comparator. These are software limits, not mathematical assumptions. Cycle mode can process much wider interfaces when the cycle rank is small. Cooperative time and operation budgets raise `ResourceLimit`; the command-line interface reports `RESOURCE_LIMIT`, not infeasibility. Checks at designated loop points are not hard real-time guarantees. The checker has its own dimension guard; reaching it means the certificate was not verified with those resources, not that the language is empty. Native dispatch must preserve this distinction and its exact fallback.

10 Further research questions and proposed work

1. Export a native source-bound envelope profile. From the existing checked exterior, identify a search producer with actual disk pieces and separated arc interfaces. Record r_i , $|\sigma_i|$, $|\rho_i|$, λ_i , types, options, and expanded witness sizes at every cut. The first target is a trustworthy source profile, not extrapolation from a raw-width statistic.

2. Prove a geometric low-overlap decomposition theorem. Which knot-exterior decompositions force most interface arcs to be tree-like in the past-future incidence graph? A constructive $O(\log^2 n)$ cycle-overlap theorem with polynomial total overhead and type control would discharge theorem 7.1. Independent cycles, rather than the mere proliferation of arcs, are now the obstruction to quantify.

3. Build a source-complete patch language for a substantial class. Prove normal matching, embeddedness, boundary-chain transport, and type substitution for a useful class of diagrams or triangulations. A restricted-class theorem with these obligations discharged is more informative than assuming all hierarchy steps fit the model.

4. Optimize orders for overlap rather than only arc count. The value $r_i + 1 - |\sigma_i| - |\rho_i|$ is computable at every cut of an explicit ordered grammar. Can representable-matroid tools find or approximate an order minimizing its maximum? Charge order discovery separately and compare against ordinary frontier-size heuristics on the same source languages.

5. Tighten envelopes without enumerating all contexts. The union-of-options compiler merges mutually exclusive choices. Can a bounded type refinement or decision diagram preserve useful alternatives? Bound the product of the number of refined types and their 2^λ dimensions, rather than improving one factor while hiding growth in the other.

6. Compress the actual suffix observation space. All refinements of ρ may greatly exceed the actual suffix language. Can backward propagation construct a smaller spanning observation family without enumerating suffix assemblies? Charge every transfer-map construction and retain independent source checking. The single-cap control demonstrates the opportunity even after this improvement.

7. Implement binary joins and product-family improvements. Extend the executable compiler and run checker to typed binary assembly DAGs. Whole-context replacement gives

the required semantic contract, but efficient products and sound envelope transport need implementation and audit. Representative sets of product families provide classical tools worth investigating [8].

8. Treat binary multiplicities without erasing cycles. A symbolic parallel-arc representation must preserve Euler corrections and every attachment cycle. Determine when it admits polynomial-bit-size cycle-space blocks. The two-arc annulus rules out replacing a bundle by a single edge. A correct theorem could address large normal weights without assuming all arcs are explicitly expanded.

9. Bind essential-boundary sectors to the source torus. Transport external boundary chains to certified cocycles with correct seam cancellations. Export a recovered witness to the maintained normal-disk verifier, and audit componentwise parities. This is necessary before treating an abstract disk outcome as an unknot certificate.

10. Extend terminal topology and permitted operations. Study bounded-genus or prescribed-boundary completion and determine the necessary defect, orientation, and boundary-order data. Circle gluing and compression surgery need a replacement for the arc-defect lemma; a small-looking algebraic state alone is not a justification.

11. Certify adaptive restrictions and recovery. Shrink future languages monotonically when possible, but preserve unpruned provenance for later widening. Prove amortized costs including envelope rebuilding and certificate storage. Choose direct scanning for short suffixes and pay basis construction only when reuse can repay it.

12. Establish geometric sharpness and formalize the finite core. Which chord-detachment families coexist in one embedded knot-region type? Which geometric restrictions force smaller ranks? Independently formalize the split-cycle identity, determinant factorization, weighted replacement, and compiler safety. These finite explicit statements are suitable first formalization targets rather than a monolithic proof of the whole recognizer.

11 Conclusion

Certified connectivity on both sides of a frontier identifies a smaller space than raw arc width: the cycle overlap of their incidence graph. Vertex splitting gives parity constraints on that space, whose maximal minors factor disk completion with exact binary rank 2^λ . Binomial grading and sharp representative-subset lower bounds hold for every connected envelope.

The polynomial compiler and whole-context proof make this compression usable in a complete explicit patch search. Independent checks and paired measurements demonstrate a local improvement together with its overhead. The sufficient quasi-polynomial condition changes from small raw width to small two-sided cycle overlap. Producing a complete source-bound geometric language with that overlap and controlled total encoded cost remains a separate research obligation, not a premise silently treated as proved.

A Artifact map and reproduction

From the extracted package directory:

```
make test
make audit
make example
make benchmark # writes new timing samples
make tables    # reads retained samples only
make pdf       # does not rerun benchmarks
make verify    # checks the supplied example run and mesh
```

The standalone command-line interface is:

```
python code/grammar.py examples/grammar.json --mode cycle \
  --seconds 30 --output results/new_run.json
```

The example resource cap is user-selected, not a promised completion time.

File	Role
article.tex, article.pdf	Proofs, analysis, and compiled article.
references.bib	Bibliographic export; the TeX has a self-contained reference list.
code/envelope_kernel.py	Envelope graph, cycle features, weighted reducer, certificates, and resource guards.
code/checker.py	Independent nullspace/minor checker, universal-graph compiler, BFS transitions, and whole-run checking.
code/grammar.py	Exact, root, and cycle finite-grammar solvers and JSON command-line interface.
code/surface_replay.py	Polygon triangulation, seam gluing, manifold, Euler, boundary, and orientation checks.
code/fixtures.py	Explicit reproducible grammars and wide-envelope fixtures.
tests/test_research.py	Forty-two test methods and negative controls.
code/audit.py,	Finite mathematical audit and paired
code/benchmark.py	complete-search measurements.
code/example.py,	Create and independently verify supplied examples.
code/verify_example.py	
results/	Executed logs, raw JSON, generated TeX tables, and CSV measurements.
examples/	Grammar, complete run certificate, surface replay, and 256-arc reduction example.
docs/CLAIMS.md,	Evidence ledger and native integration gate.
docs/INTEGRATION.md	
docs/SOURCE_AUDIT.md	Inspected sources, pin, access limits, and novelty boundary.
SHA256SUMS	File integrity manifest, not a mathematical proof.

B Minimal API example

```

from envelope_kernel import Candidate, Envelope, reduce_family
from checker import check_reduction

# Six actual arcs, one past block, four future blocks: lambda = 2.
env = Envelope((0, 0, 0, 0, 0, 0), (0, 0, 0, 1, 2, 3))
items = [Candidate(env.coordinate_partition(mask), mask - 2)
          for mask in range(4)]
answer = reduce_family(items, env)
valid, reason = check_reduction(items, env.sigma, env.rho,
                                answer.certificate)

assert valid, reason
assert len(answer.retained) == 4
# A sharp finite-interface family, not a knot certificate.

```

C Authorship, review, and source scope

These are human-readable proofs with finite computational support, not externally peer-reviewed or proof-assistant-verified results. Earlier project manuscripts and classical representative-family methods are credited. No third-party font files, copied research-paper PDFs, or upstream source archive is redistributed.

Incoming intake assigns a report number at placement. This delivery uses a descriptive name, does not select a report number, and does not modify maintained dispatch. The source audit records inspected paths and the fact that incoming ZIPs were not unpacked. Standalone sources and all retained experiment data are included, so these experiments do not depend on unrecorded background work or an unavailable native checkout.

References

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